Example 7.3

Verfy the following statement:

The mean of a Binomial distribution is 12 and its standard deviation is 4.

Solution:

Mean: $np = 12$

$$\text{SD} = \sqrt{npq} = 4$$

$$npq = 4^2 = 16 \quad , \quad \frac{np}{npq} = \frac{12}{16} = \frac{3}{4}$$

$$\Rightarrow q = 4/3 > 1$$

Since $p + q$ cannot be greater than unity, the Statement is wrong

Example 7.4

The probability that a student get the degree is 0.4 Determine the probability that out of 5 students (i) one will be graduate (ii) atleast one will be graduate

Solution:

Probability of getting a degree p= 0.4

$$\therefore \quad q = 1 - p$$
$$= 1 - 0.4$$
$$= 0.6$$

(i) P (one will be a graduate) = $P(X = 1)$

$$= 5C_1 (0.4)(0.6)^4$$
$$= 0.2592$$

(ii) P (atleast one will be a graduate)

= 1–P (none will be a graduate)

$$= 1 - 5C_0 (P^0)(q)^{5-0}$$
$$= 1 - 5C_0 (0.4)^0 (0.6)^5$$
$$= 1 - 0.0777$$
$$= 0.9222$$

Example 7.5

In tossing of a five fair coin, find the chance of getting exactly 3 heads.

Solution :

Let X be a random variable follows binomial distribution with $p = q = 1/2$

$$P \text{ (3 heads)} = 5C_x \left(1/2\right)^x \left(1/2\right)^{5-x}$$
$$= 5C_3 \left(1/2\right)^3 \left(1/2\right)^{5-3}$$
$$= 5C_3 \left(1/2\right)^5$$
$$= 5/16$$

Example 7.6

The mean of Binomials distribution is 20 and standard deviation is 4. Find the parameters of the distribution.

Solution

The parameters of Binomial distribution are n and p

For Binomial distribution Mean = np =20

Standard deviation = $\sqrt{npq} = 4$

$$\therefore \quad npq = 16$$

$$\Rightarrow \quad npq/np = 16/20 = 4/5$$

$$q = 4/5$$

$$\Rightarrow \quad p = 1 - q = 1 - (4/5) = 1/5$$

Since $np = 20$

$$n = \frac{20}{p}$$

$$n = 100$$

Example 7.7

If x is a binomially distributed random variable with $E(x)$ =2 and van $(x) = 4/3$. Find $P(x = 5)$

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